DESN30036 Interiors 6

PROJECT 1

RESEARCH COMPONENT





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Jan 2024



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Research Article & LTC Manual

"Biophilic Design as an Important Bridge for Sustainable Interaction between Humans and the Environment"

According to the research Biophilic Design as an Important Bridge for Sustainable Interaction between Humans and the Environment: Based on Practice in Chinese Healthcare Space，" The healthcare spaces have become one of the most visible venues for the green building movement as a result of concerns about hospitals' environmental implications(Zhao, Zhan, & Xu, 2022)". The article suggests a variety of psychological and cognitive benefits of biophilic design, such as positive effects on patients' physiology and psychology, reduction of negative emotions such as panic and anxiety, stimulation of patients' physical potential to promote health recovery, and reduction of employee stress and fatigue.

The biophilic design methods are summarized in the article:

The biophilic design methods from Stephen Kellert in 2015 [29].

Classification of design methods	Design elements
Direct nature experience	(1) light; (2) air; (3) water; (4) plants; (5) animals; (6) weather; (7) natural systems; (8) fire
Indirect nature experience	(9) natural patterns; (10) natural materials; (11) natural colors; (12) simulated natural light and natural ventilation; (13) natural shapes or forms; (14) natural associations; (15) information richness; (16) change of time; (17) natural geometry; (18) bionics
The experience of space and place	(19) foresight-shelter; (20) organize complexity; (21) integration; (22) transitional spaces; (23) mobility and wayfinding; (24) emotional connection of place

Zhao, Zhan, & Xu, 2022

Research Article & LTC Manual

Based on research and site visits to 12 hospitals, the researchers made several recommendations regarding design. First, windows are needed to provide natural light. According to the Ontario Long-Term Care Design Manual, Bedrooms must have at least one window that provides a direct view to the outdoor. The proper windows expand the view of the natural landscape outside the room and helps increase the amount of light. At the same time, LTC also specifies that windows need to be shaded during certain seasons. One of the hospitals uses bamboo and greenery to create shadows, and large windows are also used to increase natural light. This is a design that helps to communicate between the indoor and outdoor spaces and increases the occupants' connection to nature.

Warm walls, the use of landscape paintings, the placement of ornamental fish, and the use of natural materials in the furniture are all listed in the article as biophilic designs. Ornamental fish serve as direct nature-experiencing element, providing the occupants with the opportunity to interact with living organisms. The rest are non-nature-experiencing elements that enhance their connection to nature and provide them with benefits more psychologically and perceptually.

The researchers conducted a statistical survey to assess the benefits of using each element. The most important of the conclusions were "creating a relatively inward-looking spatial environment, similar to a cave in nature, with long, open views in the foreground", "using real plants or specimens, green roofs, green walls", and "using natural construction or decorative materials" (Zhao, Zhan, & Xu, 2022). From the results of the public's assessment of the importance of biophilic design elements, it can be concluded that they need small spaces that provide shelter and prefer direct natural elements.

There are different needs for different groups of people. For example, the study included a number of healthcare facilities that serve children. In these places, designs using bright colors and cartoonish nature-related motifs were given positive ratings.

The article also mentions methods of improving the interior such as adding gardens, adding skylights, and introducing natural elements. At the same time, it emphasizes the importance of a biophilic design process that carefully selects natural elements and employs a highly controlled design to avoid possible risks and burdens (Zhao, Zhan, & Xu, 2022).

Building Case Study: Östra Hospital

Östra Hospital is an example of a health care facility with biophilic design. The project is a psychiatric facility in Gothenburg, Sweden, designed by White Architects. "The designers believed that if the facility was built to be inescapable, it would encourage escape attempts. If it is built to be beautiful and welcoming, it is more likely to be respected and contribute to the well-being of its occupants" (Nestor, 2017).



Nestor，2017



Nestor，2017

Östra's floor plan consists of four repeating L-shaped sectors, three outdoor courtyards and a connecting pathway. Each department is self-contained and includes treatment rooms, administrative offices and living areas. This repetitive arrangement of spaces allows staff to easily navigate the entire facility once they are familiar with one department.

The design also ensures that all patients have continuous visual access to natural elements. Through the layout of the three central courtyards, people can enjoy the natural landscape even from their rooms in the center of the building. This design enhances the indoor experience and encourages patients to get outside. It also makes it easy for staff to monitor the movements of all patients.

Building Case Study: Östra Hospital

Dynamic and diffused natural light is found throughout the facility. This variation of light creates the effect of light and shadow in natural conditions, which can be beneficial for depression, sleep disorders and more. Each patient room is equipped with a corner sunroom with windows that are glass on both sides. Private rooms are equipped with windows, which are designed to contribute to physical and mental health.



Nestor，2017

Research has shown that refuge spaces can reduce stress and allow patients to feel in control of their surroundings. The design is filled with examples of refuge, from individual rooms to shared sunrooms and gathering spaces to central spaces with light courtyards. For example, the shared sunroom features operable sunshades that block outside views and increase the sense of refuge (Nestor, 2017).

The design of the Östra Hospital is very much in line with the LTC design manual for patient accommodation. According to the LTC manual, bedrooms should not accommodate more than two beds, and Östra Hospital meets this requirement by specializing in single or double rooms that also foster a sense of privacy and personal space. Each room in the facility is a self-contained unit with an in-suite bathroom, thereby increasing convenience and reducing risk. In addition, the window architecture in each room promotes a visual connection to the outdoors, ensuring an ample influx of natural light, which helps regulate circadian rhythms and promotes wellness. The facility's careful layout and generous open spaces not only ensure accessibility for all patients, but also fulfill safety and security needs. Staff monitoring is simplified by enabling clear lines of sight, which harmonizes design principles with the operational needs of the psychiatric care environment.

Equipment, Plumbing & Furniture

Some common health care medical equipment needs include:

- **Bathroom safety equipment:** Grab bars, lifts, mats, shower chairs, raised toilets, tub bars, walk in bathtubs.
- Beds and bed rails: Adjustable beds, hospital beds, bed safety rails
- **Ceiling lifts:** Patient lifts and transfer poles to help individuals get in and out of bed
- **Ramps:** Ramps to assist those in wheelchairs. Ramp type will depend on the entry way.
- **Commodes:** Stationary, wheeled and shower commodes
- **Lift chairs:** Lift chairs to make it easier for individuals to get in and out of chairs.
- **Wheelchairs:** Manual wheelchairs to help residents get around. The type of chairs available will depend on the facilities.
- **Stair lifts or elevators:** To assist residents to go up and down stairs and have access to different levels of the facility.

(MED+, 2020)



Patient Bed, Speciality Furniture, 2014

Stair Lift, MED+, 2020



- Long-term care home's should all ensure that all plumbing fixtures with hose attachments are equipped with backflow devices. (Ontario Regulation, 2022)

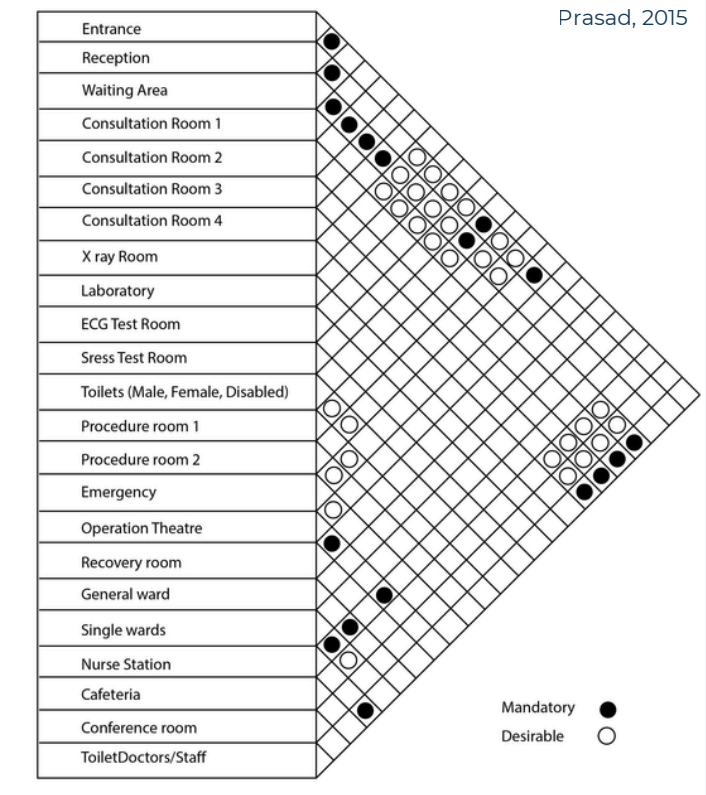
Some of the furniture used in patient rooms, public areas include:

- **Patient Room Furniture:** Beds, headboards, footboards, side rails, mattresses, nightstands, bedside tables and chairs.
- **Seating:** A variety of seating options such as recliners, rocking chairs, ottomans, benches, and handicapped seating.
- **Outdoor Furniture:** Durable and comfortable outdoor seating, tables and accessories, as well as cushions and umbrella fabrics.
- **Tables:** Includes occasional and specialty tables with pedestals and four-legged bases for different functional needs.
- **Accessories:** Bookcases, TV cabinets/shelves, fireplaces and lighting options to enhance the comfort and utility of interior spaces.

(Speciality Furniture, 2014)

Adjacencies & Natural Light

By analyzing an existing adjacency matrix, the possibilities of proximity between different rooms in a healthcare facility can be obtained.



Prasad, 2015

- Entrances should have direct access to key areas such as the front desk, waiting area, emergency room and restrooms. Critical functional spaces such as operating rooms, recovery rooms, and emergency rooms should be very close together to improve the efficiency of patient care.

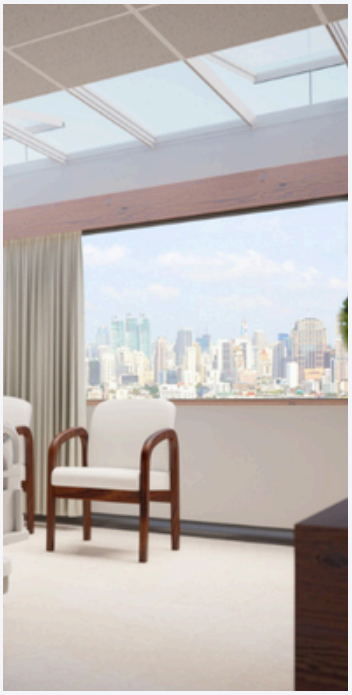
- Counseling rooms should be easily accessible from waiting areas. Laboratories, X-ray rooms, EKG rooms, and stress test rooms should be reasonably close to counseling rooms. Nurse stations should be centrally located in general and single patient rooms for effective monitoring and quick response to patient needs.

- Natural light is critical to hospital environments, regulating a person's biorhythms and affecting emotional well-being, heart function, body temperature.

- Studies have shown that patients who receive adequate natural light have a 41% shorter hospital stay and a lower mortality rate than those in dimly lit or windowless rooms.In addition, natural light accelerates post-operative recovery, reduces the need for pain medication by about 22 %, and boosts healthcare staff morale and productivity.

- Ideal hospital design should include plenty of natural light, with windows comprising at least 25 percent of the exterior walls and providing at least 2,500 lux of bright light.

(Wiseman, 2021)


Wiseman, 2021

Privacy, Acoustics & Social Organization

- Space planning in healthcare environments should prioritize the patient's acoustic experience and privacy needs. Designers need to consider current and future noise sources that may affect the facility. Designs should incorporate noise mitigation strategies such as the use of barriers, acoustic blinds, and the selection of quieter equipment to ensure a rehabilitative environment for patients.

- Single-bedded rooms reduce infection risk and improve sound insulation. Decentralized nurse stations can reduce hallway noise and enhance patient observation. The design should also incorporate private spaces to create a peaceful and respectful hospital atmosphere.

- Soundproofing goes beyond space planning and includes the choice of building materials. The overall acoustic design should consider the composite acoustic performance of walls, ceilings, doors and floors to ensure privacy and minimize noise transmission. For example, acoustic ceiling tiles are effective in reducing reverberation and improving speech intelligibility.

(Acoustics In Healthcare Environments, 2010)



Acoustics In Healthcare Environments, 2010

- The concept of social organization can be helpful in understanding the mechanisms by which a healthcare system operates.

- The design should focus on the patient experience and needs. This includes creating comfortable waiting areas, private and serene treatment rooms, and considering patient navigation through the healthcare environment. Taking into account the division of labor and collaboration between different professions in healthcare, the design should provide spaces that facilitate communication and collaboration among cross-professional teams. For example, design open office areas to facilitate the flow of information and collaboration among doctors, nurses, and technicians.

- The design should encourage effective communication between patients and healthcare professionals. For example, consultation rooms should have appropriate seating and table layouts to promote face-to-face conversations.

- Healthcare needs change over time, so designs should be flexible. Moveable partitions, multifunctional furniture, and adjustable lighting systems can be used to accommodate different treatment needs and types of activities.

(Griffiths, 2003)

Accessibility& Code Requirements

By summarizing the LTC manual, here are some parts of the interior design to keep in mind:

- Each RHA shall be a clearly defined individual unit, located on the same floor, for a maximum of 32 residents. Resident spaces (including bedrooms, bathrooms, dining areas, lounge and activity areas) and staff work areas (including records areas, treatment spaces, equipment and supply storage and utility spaces) shall be located within each RHA.

- Each bedroom shall have a maximum of two beds and a minimum of 12.08 square meters (130 square feet) of usable net floor space for single bedrooms and a minimum of 10.68 square meters (115 square feet) of usable net floor space per resident is required for double bedrooms. Bedrooms must have en-suite bathrooms with a minimum bathroom doorway width of 914 mm (3 ft.) and sufficient turnaround space to accommodate a wheelchair or walker.

- Emergency call systems, and locks that are easy for residents and staff to open from the outside near the bed and toilet are required.

- Each RHA must have a workspace for caregivers to do their work. There must be a space on each floor of the RHA that supports the therapy program. At least one therapy room is required for every three RHAs.

- Each RHA must have space to store needed medications, supplies, and equipment that should be easily accessible to caregivers without intruding into residents' personal space.

Accessibility& Code Requirements

- Outdoor space must be provided on each floor of the RHA and must be fenced. The outdoor space must not be more than 61 meters (200 feet) from the farthest resident bedroom. Enclosed outdoor space railings/fences/gates must be at least 1.6 meters (5 feet) in height.

- Landscaping and design of outdoor spaces must take into account the safety needs of residents. Outdoor spaces accessible to residents should include hard, flat surfaces and should not include slopes and steps. Each outdoor space must have areas that provide shade, seating, and protection from wind and other weather elements.

- Corridors in all residential areas must be at least 1820 mm (72 inches) wide. Sturdy handrails must be installed on both sides of the corridor walls. Handrails shall be installed at a minimum height of 860 mm (34 inches) above the floor.

- Each RHA must have easily accessible clean and contaminated utility rooms to support the storage, cleaning and sanitizing of resident care equipment.

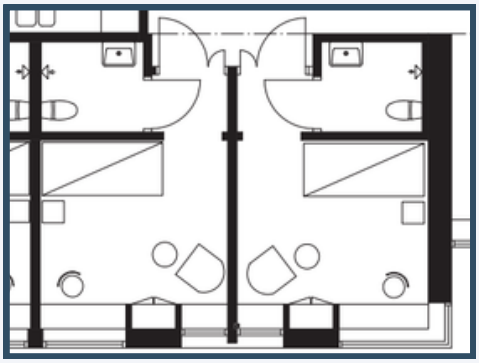
- There must be a secure area, inaccessible to residents, for environmental controls and other building system controls.

- The minimum total lounge and activity space requirement per resident is 2.5 square meters (27 square feet). At least 70% of the required lounge and activity space must be located inside the RHA. The remaining space may be located outside the RHA for use by all residents of the long-term care residence.

- At least one resident lounge of no less than 14 square meters (150 square feet) and one activity space of the same size are required. At least one window facing directly outdoors is required, as well as a restroom separate from the resident's bedroom or bathroom.

Private & Public Rooms

Picture



Nestor，2017



Nestor，2017



Nestor，2017



Nestor，2017

Description

The figure shows the floor plan of two adjacent patient rooms in Östra Hospital. It can be seen that both are single patient rooms with separate bathrooms. Sufficient chairs and tables were provided. The windows can be opened, but they are not opened too wide, which is also in accordance with the LTC manual.

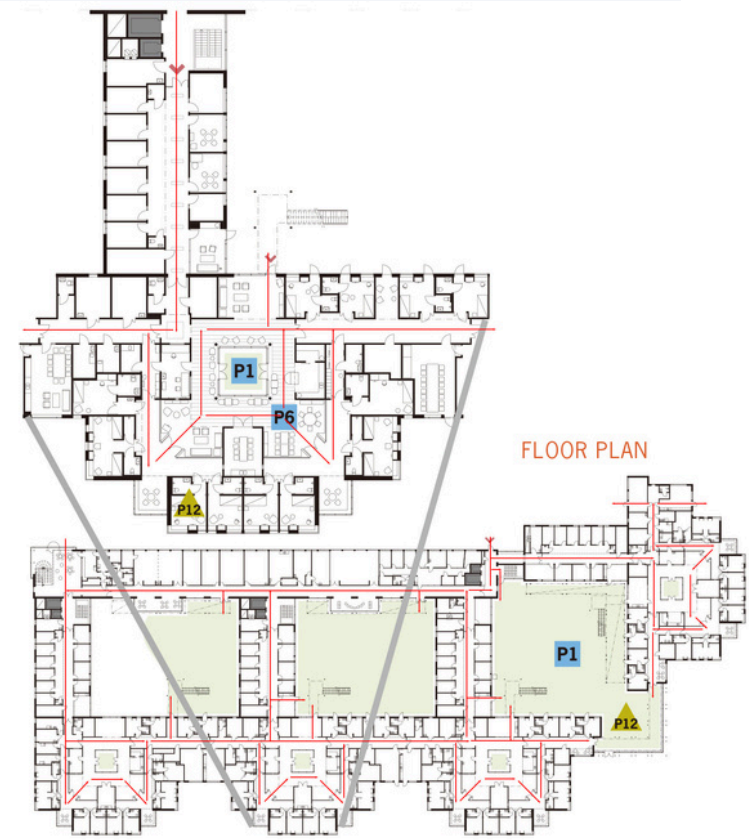
The hospital provides plenty of natural light and lounging spaces close to nature. With this small greenhouse as the centerpiece, wards, service spaces and dining areas are evenly distributed so that each room is adjacent to natural spaces. At the same time, this plan allows for good visibility and makes it easy for healthcare workers to care for all patients.

These outdoor spaces have been carefully designed and planted with a variety of plants to create a peaceful environment. Ample natural light in these spaces further enhances the therapeutic effects. This environment helps patients reduce stress, promote relaxation and improve mood. In addition, these outdoor areas provide an ideal environment for activities.

This is a multi-purpose room that can be used for counseling. Shades help to block out the sunlight and diffuse light into the room, which helps to create a relaxing and welcoming environment that relaxes the patient's psychological stress. Many potted plants have been placed in the room to provide this area with direct contact with nature.

Circulation & Wayfinding

- Circulation at Östra Hospital is straightforward and efficient. The four main areas are laid out in a similar way and have good visibility, making it easy for patients and staff to find their way around.
- The design ensures that all facilities are easily accessible and that there are no dead ends.
- Numerous doors to the outside world have been provided, which ensures safe evacuation and at the same time gives patients easy access to the natural environment.



The hospital does not have a very obvious wayfinding system and could be improved based on the following:

- Place clear, simple signage at key points, such as entrances, corridors, elevators, stairways and patient rooms. Unique artwork, architectural features or color schemes can be utilized as orientation and memory aids. Ease navigation through systematic layouts and numbering, such as wing layouts and route numbering.
- Digital wayfinding tools such as interactive kiosks, mobile apps, and touch screens can be used to provide real-time information and personalized navigation. Use tools such as social media platforms and chatbots to help users plan visits and access information.
- Hospital staff need to be familiar with the signage system and be able to effectively help visitors navigate. Design with all users in mind, including those with mobility, vision or hearing limitations, for example by using high-contrast fonts, tactile elements and Braille.

(Essential Design Principles for Hospital Wayfinding, 2023)

Healthcare Finishes



Porcelain tile

- Shades 2.0: Thunder
- Company Name: Crossville Studio
- Produced in USA
- 24"x24", Semi-polished

https://www.crossvillestudios.com/Products/CRO-SHADES-2-0#documents

- Can be applied to counters, exterior walls, interior floors and walls, pool fountain waterline.

- This tile has a wide range of applications and can be used both indoors and outdoors and in wet areas. It is hard-wearing and durable, and can withstand the intense use of healthcare spaces. It is waterproof and resistant to chemicals, which prevents liquids such as sterilizers from damaging the surface or leaving a residue in the healthcare space.



Resilient Wall Base

- Wall Base
- Company Name: Armstrong Flooring
- Composition: Thermoplastic Rubber
- Coved Wall Base - 0.125 in. (3.18 mm) Sculptured Wall Base - 0.25 in. (6.35 mm)

https://www.armstrongflooring.com/commercial/en-us/products/ima/wall-bases.html

- This wall base has a long lifespan and has durable and easy-to-clean qualities that make it suitable in healthcare spaces. It has been tested to meet standards for light and chemical resistance, fire resistance, and low VOCs. This not only ensures that it is suitable for use in healthcare environments that are exposed to long periods of light and are frequently cleaned, but also avoids the health hazards that voc poses to staff and patients.

Healthcare Finishes



Paint

- Eco Spec® Interior Paint
- Company Name: Benjamin Moore
- Finish: Flat, Eggshell, Pearl, Semi-gloss
- 3.79L (400sqft)

https://shop.thepaintpeople.com/products/eco-spec%C2%AE-zero-voc-zero-emission-waterborne-interior-paint

- This paint is suitable for commercial, facility management and residential applications.

- It has zero VOC, zero emission and low odor qualities. It has green seal, anti-asthma, anti-allergy and anti-mold certifications. All of these features are important in healthcare environments. Zero emissions ensure indoor air quality, and the anti-asthma and anti-allergy features take care of special populations. Acrylic paints are water resistant and paired with anti-mold features for long lifespan in healthcare environments.



Horizontal surface materials

- Compact Laminate
- Company Name: Wilsonart Engineered surfaces
- Thickness from 1/10" to 1"
- A variety of colors and finishes

https://www.wilsonart.com/compact-laminate#

- Can be applied to lockers and cabinets, elevator cabs, wall panels, casework, shelving, specialty furniture, fixtures and partitions.
- More robust and durable than traditional laminates, this material has an easy-to-clean surface with excellent chemical and stain resistance for healthcare environments. According to data provided, it is fire-resistant and can also withstand boiling water immersion, has good dimensional stability, and maintains its shape under temperature and humidity changes. It is also available in a wide range of colors, which can be used as part of a wayfinding system or be noticed by colorblind people.

Healthcare Finishes



Ceiling Tiles

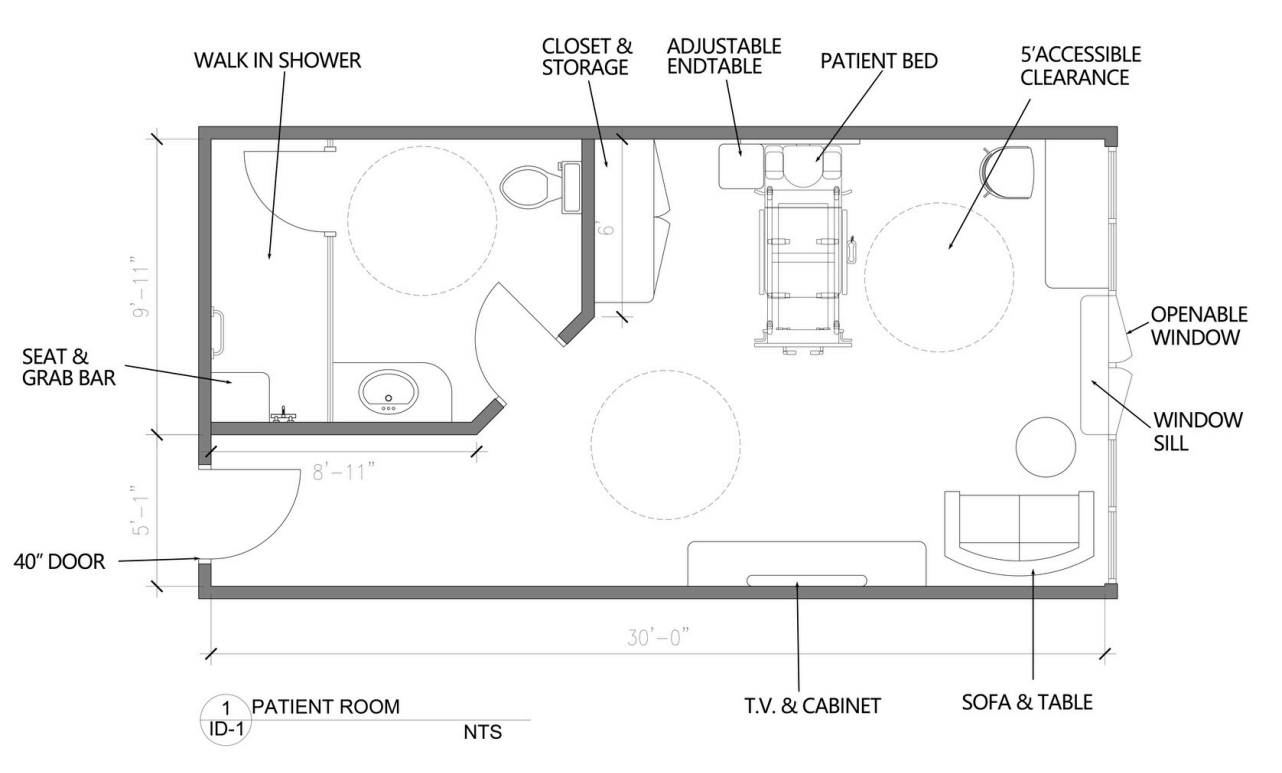
- Rockfon® Medical™ Air Acoustic ceiling tile
- Company Name: Rockfon
- 24"x24"x1" or 24"x48"x1"

https://www.rockfon.com/products/rockfon-medical-air/#ProductOverview

- Can be applied to intensive care，operation room，recovery room and laboratory.

- This ceiling tile offers excellent fire safety ratings, high sustainability with a large number of recycled content, enhanced surface durability for easy cleaning, and resistance to common cleaning chemicals. Superior hygiene properties tested against persistent microorganisms and effective air pressure control features make it suitable for healthcare environments.

A Typical Individual Patient Room



- Firstly, exposure to natural light improves the mental and physical health of patients, and a large number of glass windows have been set up to achieve this. In the center, an openable window is set up, which can be opened at an angle that is not too large, depending on the requirements. A window sill is provided for the patients to use or to place green plants to enhance the patients' contact with nature.
- To avoid the bathroom obscuring the window area, as well as to provide privacy, it is set in the upper left corner. An angled door avoids the oppressive feeling of the corridor and facilitates patient access. The interior of the bathroom is accessible, and the walk-in shower has a seat and handles in it, which are safety considerations.
- A closet and adjustable end table are placed at the foot of the bed for easy access to the patient's personal belongings. TVs and cabinets across from the bed provide entertainment and storage space without interfering with the main walkway, and they are positioned so that patients can watch TV from the comfort of their bed.
- Beds can be reached from both sides for easy access to caregivers. The spacious corridors are also easy to use and access for special populations.

Conclusion

- The following healthcare interior design recommendations can be gleaned from research:

- Ensure rooms have enough natural light and provide a visual connection between indoors and outdoors through large windows or sunrooms to promote patients' mental and physical well-being. Use natural paintings as well as furniture made of natural materials to enhance the patient's connection with nature and boost psychological feelings. Use appropriate shading during different seasons to ensure that sunlight enters the room, but also to avoid excessive sunlight. At the same time, the use of dynamic and dispersed natural light to mimic the natural environment contributes to the emotional and physical well-being of patients.

- Provide patients with private resting spaces, such as single or double occupancy rooms, to reduce stress and enhance a sense of security. Ensure rooms have emergency call systems. Consider the patient's acoustic experience and privacy needs in the design and use soundproofing strategies to create a quiet therapeutic environment. Ensure the design is flexible to accommodate different treatment needs and activity types, including multi-functional furniture and adjustable lighting systems.

- Implement clear navigation systems in healthcare facilities, including clear and simple signage, digital navigation tools and easily recognizable navigational elements to help patients, visitors and staff find target locations more easily.

- In terms of material selection, choose hygiene-compliant interior surfaces, such as antimicrobial materials, easy-to-clean surfaces, and products that are fire-resistant and low in VOCs to ensure a sanitary and safe healthcare environment.

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